

Fronthaul/Midhaul Networks: Capacity and Latency Requirements Imposed by 6G Disaggregated RANs

Ana Larrañaga, Sandra Lagén, Josep M. Fàbrega, José Manuel Rivas-Moscó, Juan Pedro Fernández-Palacios, Ioannis Tomkos, and Raül Muñoz

ABSTRACT

This article analyzes the capacity and latency requirements imposed by various 6G envisioned scenarios and technologies over the fronthaul/midhaul networks when considering radio access network (RAN) disaggregation. We also examine how emerging 6G technologies — such as multi-connectivity, coordinated multi-point, and relays — impact on these requirements. For the capacity requirements, we differentiate between basic and advanced 6G RAN scenarios, considering both single-carrier and multi-carrier setups, while extrapolating configuration parameters based on previous generation systems. For the latency requirements, our study is based on values already envisioned for 5G Advanced. Finally, we propose various transport network scenarios tailored to address such requirements in future 6G fronthaul/midhaul networks, focusing on two specific use cases: flexible function split (FFS) and cell-free multiple-input multiple-output (MIMO).

INTRODUCTION

Currently, the Sixth-Generation (6G) wireless network is undergoing initial definition of its targeted use-cases and capabilities [1]. Early works suggest that 6G will introduce advancements in several use-cases/applications, including ultrahigh data rate transmissions for holographic images, ultra-low latency communications for autonomous driving and industrial production, and ultra-dense connectivity to support crowded environments such as shopping malls [2]. These emerging applications will be supported by a new era of wireless communication where innovative technologies will be needed.

The Fifth-Generation (5G) network has introduced various advancements to enhance flexibility and efficiency within mobile networks. The latest development includes the virtualized base station, or gNB, which facilitates the disaggregation of the RAN into different units. This architecture allows various RAN functions to operate independently in different physical baseband entities, rather than relying on a single, closed, and monolithic system. As a result, components such

as the data processing unit in the Physical (PHY) layer, the radio, or the Medium Access Control (MAC) layer can be separated and provided by different vendors.

The disaggregated RAN comprises three components called the Radio Unit (RU), the Distributed Unit (DU), and the Centralized Unit (CU), which are placed across different geographical locations. The RU-DU and the DU-CU are connected via transport networks (or dark fiber), called fronthaul and midhaul networks, respectively. Through virtualization, functional splitting becomes feasible and determines the location of protocol layer functions across each unit. The 3rd Generation Partnership Project (3GPP) specifications have defined functional split Option 2 for CU/DU split but has also explored the trade-offs of other split options [3]. Conversely, the Open RAN (O-RAN) Alliance has introduced the new RU entity and the open fronthaul interface, while adopting functional split Option 7.2x (intra-PHY split) for the DU/RU division [4]. Based on the chosen functional split, the transport network capacity and latency requirements vary [3].

We assume that 6G will inherit the benefits of using functional splits and RAN disaggregation, which enhance efficiency in terms of energy consumption [5]. However, as the 6G standard will evolve in comparison to 5G, there is a need to establish novel requirements for transport networks tailored to the unique characteristics of the 6G network. This is crucial to facilitate the operation of functional splits while fully exploiting the potential of 6G. Furthermore, knowing these requirements is key to design new transport network technologies for future 6G fronthaul/midhaul networks.

In this context, this article explains the process of calculating transport network capacity and latency requirements for various functional split options in 5G networks and extends state-of-the-art computations to determine the new requirements that the transport network must meet for future disaggregated 6G networks. To achieve this, first, we explain basic RAN concepts and

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the envisioned enabling technologies for 6G networks. Then, we define four 6G RAN scenarios based on state-of-the-art specifications and conservative extrapolations. Moreover, to support the new requirements imposed by 6G on the transport network, optical technology and architecture advancements are crucial [6]. Given these requirements, this article proposes potential 6G fronthaul/midhaul transport network scenarios tailored to two distinct 6G use cases (i.e., FFS and cell-free MIMO), aiming to fulfill their anticipated transport network needs.

BASIC CONCEPTS AND 6G ENABLING TECHNOLOGIES

Functional splitting is possible thanks to RAN disaggregation and virtualization, which determines the placement of protocol layer functions across different units. To comprehend the requirements that the transport network must meet, it is essential to understand the processes occurring within each protocol layer, influenced by the technologies in use, such as MIMO, along with the information and signaling exchanges between these layers, which may be managed by different units. As 6G has not been standardized yet, the technologies that will be used are not defined. However, numerous state-of-the-art studies, along with the latest 3GPP releases, are exploring emerging technologies for the 6G RAN [7, 8]. These technologies may impact the transport network requirements. In this section, we introduce the basic concepts of MAC and PHY layers in order to list the most important parameters and processes to consider in these protocol layers and additionally, we discuss some technologies currently being studied as potential candidates for 6G RAN.

BASIC CONCEPTS

The MAC layer is responsible for assigning radio resources for data packet transmission of the various multiple Data Radio Bearers (DRB) *s* across the air interface. Through a scheduling mechanism, it allocates time and frequency radio resources and performs adaptive modulation and coding for each packet/DRB/user while preventing collisions between assignments. This information, along with user data, is sent to the PHY layer (in the Downlink (DL) case), to be subsequently transmitted to the UE through the wireless network. Furthermore, the MAC layer is also responsible for the Hybrid Automatic Repeat reQuest (HARQ) process, which handles retransmissions of unsuccessfully received packets by the User (UE), including the HARQ feedback processing and radio resource reallocation.

When the data packet is received in the PHY layer, channel coding is initially applied. This involves adding redundancy bits to facilitate error detection and correction at the receiver side [9]. The resulting output, a codeword, is multiplied by a scrambling sequence to mitigate the same frequency signal interferences. The scrambled codeword bits are mapped into complex modulation symbols according to the Modulation and Coding Scheme (MCS) chosen by the MAC layer. The chosen MCS determines the Modulation order (M) and the effective code rate that is applied. 5G uses MIMO technology to perform multi-stream transmissions, that is, send multiple data streams in the same time/frequency resourc-

es. In the PHY layer, the modulated symbols are mapped to N_S spatial streams, and then combined and redistributed over several antenna ports (N_{AP}). These antenna ports are mapped onto one or more physical antenna elements. An antenna port represents a logical concept in which each port generates a resource grid. Then, considering the decision taken by the MAC scheduler, the modulated symbols that have to be transmitted through each antenna are mapped to available resource elements (i.e., 1 subcarrier in the frequency domain and 1 Orthogonal Frequency Division Multiplexing (OFDM) symbol in the time domain). For each subcarrier, the Inphase and Quadrature (IQ) samples are obtained. The cyclic prefix is added to these IQ samples, and they are converted from digital to analog in order to be transmitted through the wireless channel, following a specific Sampling Rate (SR). Later we will see how these parameters are extended for 6G.

6G RAN ENABLING TECHNOLOGIES

The articles [7] and [8] summarize the technologies being considered for enabling 6G networks. Regarding multiconnectivity, one key feature is carrier aggregation, where the UE can combine multiple independent Component Carriers (CCs), allowing access to the gNB with a larger aggregated bandwidth. Additionally, dual connectivity allows a UE to access different RUs simultaneously — one master and one secondary — improving both reliability and throughput. With the coordinated multi-Transmission and Reception Point (TRP) technologies, joint transmission is highlighted, where the same data is transmitted simultaneously to the UE by multiple coordinated RUs acting as a single spatially distributed transmitter. There is also dynamic point selection, which enables user data to be transmitted from different RUs at different times, depending on channel conditions and resource availability. Another important aspect is Uplink (UL) and DL decoupling, which allows DL/UL transmissions to be managed by different RUs, and there will be another base station to manage the control messages. Furthermore, the UE can function as a relay to extend network coverage, retransmitting data to another UE that is outside the coverage of the gNB. In case of NR repeater, it is a repeater with beam-forming capabilities and under network control. Concepts of intelligent reflective surfaces are also being explored for passive repeaters.

The impact of these technologies on transport requirements must be studied separately. Regarding capacity, when carrier aggregation is considered, the aggregated bandwidth in the air increases, thus increasing the traffic transmitted by the fronthaul of a given RU. In the rest of the technologies, as each RU has its own fronthaul, the fronthaul capacity calculation is not affected. Regarding latency, in the case of dual connectivity, joint transmission, and dynamic point selection, latency will be determined by the maximum transmission time between the multiple RUs and the UE. Furthermore, for NR repeaters or UE-to-network relays, the latency experienced in the fronthaul will be more restrictive, as the signal will need to be transmitted to an intermediate node (the relay or repeater), where the data will be processed before being forwarded.

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	Reference 4G	Reference 5G	6G Basic	6G Advanced
Peak Rate (PR)	DL: 150 Mb/s, UL: 50 Mb/s	—	—	—
Channel bandwidth (BW)	DL/UL: 20 MHz	DL/UL: 100 MHz	DL/UL: 400 MHz	DL/UL: 1 GHz
Modulation (M)	DL: 64-QAM (M = 6), UL: 16-QAM (M = 4)	DL: 256-QAM (M = 8), UL: 64-QAM (M = 6)	DL/UL: 256-QAM (M = 8)	DL: 1024-QAM (M = 10), UL: 256-QAM (M = 8)
Number of MIMO streams (N_S)	DL: 2, UL: 1	DL/UL: 8	DL/UL: 8	DL/UL: 16
Number of antenna ports (N_{AP})	DL/UL: 8	DL/UL: 32	DL/UL: 64	DL/UL: 256
Sample rate (SR)	—	DL/UL: 30.72 Msamples/s	DL/UL: 4×30.72 Msamples/s	DL/UL: 10×30.72 Msamples/s

TABLE 1. Assumed 6G network configuration reference values for DL/UL in 6G basic and 6G advanced scenarios.

ENVISIONED 6G RAN SCENARIOS

Toward foreseeing 6G transport capacity requirements, we define four 6G RAN scenarios: basic and advanced 6G scenarios with and without multi-band (or carrier aggregation) capabilities. In the conservative or basic scenario, we consider the values of 5G Advanced scenarios for massive MIMO use cases. For the 6G advanced scenario, we extrapolate the values for 6G massive MIMO based on scaling factors derived from previous generation systems (3G, 4G, 5G). Moreover, we explore the single-band and multi-band configurations with multiple CC for both scenarios, as adopted in 3GPP, which significantly increases the aggregated bandwidth that can be used in the air interface, and thus the associated required transport capacity. Table 1 shows the considered 4G reference, 5G reference, 6G basic and 6G advanced singleband scenarios, considering DL and UL configurations.

We assume a Channel Bandwidth (BW) for 6G basic of 400 MHz for DL and UL. This BW is already considered in frequency range 2 (FR2, covering carrier frequencies spanning from 24.25 GHz to 71.0 GHz) of 5G NR. For the 6G advanced scenario, we take into account the most recent 3GPP TS 38.101-2 (release 18), which already considers a BW of 2 GHz for data transmission. Therefore, we assume that a BW of 1 GHz will be generic in the 6G advanced network. In scenarios involving multi-band configurations, 5G currently supports transmissions across 16 CCs [10]. Therefore, extrapolating from a basic 6G bandwidth of 400 MHz for a single CC, the aggregated bandwidth for 16 CCs would be 6.4 GHz. For the case of a 6G advanced multi-band network, we observe that 4G currently supports 5 CCs, whereas 5G expands this capability significantly by combining up to 16 CCs. This represents a threefold increase in the number of supported CCs from 4G to 5G. By extrapolating this trend, 6G advanced networks will further elevate this capacity, enabling the aggregation of 48 CCs or even more, thus resulting in 48 GHz bandwidth.

In the case of modulation, 5G supports (in the latest releases) up to 1024-QAM in DL (i.e., $M = 10$) and 256-QAM in UL ($M = 8$). However, following the reference values in [3], we consider 256-QAM a generic modulation for 6G basic, and 1024-QAM for 6G advanced in the DL.

In order to configure the number of MIMO streams and antenna ports we have used latest 3GPP 5G NR values for 6G basic and considered the most recent advancements provided by Qual-

comm and Ericsson for 6G advanced. On the one hand, for the 6G basic scenario, we keep the number of MIMO streams in 5G, that is, 8 streams. For 6G advanced, we consider a conservative extrapolation to 16 streams, for both DL and UL (note that 4G used 2 streams for DL and 1 for UL, and 5G used 8 streams for both DL and UL). On the other hand, for antenna ports, Ericsson has already developed 64Tx64Rx equipment and Qualcomm has equipment operating in the 3.5 GHz band. Considering the current commercial situation, we assume that 6G basic will use 64 antenna ports for DL/UL transmissions. Moreover, 256 antenna ports have already been considered in [3], and Qualcomm has already a prototype for a 6G antenna array that operates in the 13 GHz band with 4096 antenna elements and 256 antenna ports. So, we adopt 256 antenna ports for the 6G advanced scenario.

Finally, in terms of sampling rate, we assume the same proportional scaling as in the BW case. A typical 5G sampling rate is equal to 30.72 Msamples/s. Then, for the 6G basic scenario, we increased the sampling rate by a factor of 4 (from 100 MHz to 400 MHz), while for the 6G advanced scenario, we multiplied the sampling rate by a factor of 10 (from 100 MHz to 1 GHz). For the multi-band configuration, we considered 16 CCs for 6G basic and 48 CCs for 6G advanced. This results in scaled sampling rates of $4 \times 16 \times 30.72$ Msamples/s and $10 \times 48 \times 30.72$ Msamples/s, respectively.

TRANSPORT NETWORK REQUIREMENTS FOR 6G

The transport network capacity and latency requirements for each functional split are computed based on the assumed RAN scenarios for the 6G network. These requirements apply to the transport link in which the functional split is adopted. If the split is performed between the CU and the DU, these units communicate through the midhaul network. Conversely, if the split is applied between the DU and the RU, these units communicate through the fronthaul network. Therefore, the derived requirements apply, respectively, to each network segment. In this section, we explain the capacity and latency calculations for different functional splits, as well as the required values considering the RAN scenario configurations described earlier.

FRONTHAUL/MIDHAUL CAPACITY REQUIREMENTS

With functional split 1 (Radio Resource Control (RRC)), split 2 (RRC and Packet Data Convergence Protocol (PDCP)), and split 4 (RRC, PDCP and Radio Link Control (RLC)) high protocol lay-

Split	Capacity	6G Basic		6G Advanced	
		DL	UL	DL	UL
1	$PR \cdot \frac{BW}{BW_{ref}} \cdot \frac{N_s}{N_{s_{ref}}} \cdot \frac{M}{M_{ref}}$	16 Gb/s	16 Gb/s	100 Gb/s	80 Gb/s
2	$PR \cdot \frac{BW}{BW_{ref}} \cdot \frac{N_s}{N_{s_{ref}}} \cdot \frac{M}{M_{ref}} + S_{PDCP}$	16.01 Gb/s	16.024 Gb/s	100.02 Gb/s	80.024 Gb/s
4	$PR \cdot \frac{BW}{BW_{ref}} \cdot \frac{N_s}{N_{s_{ref}}} \cdot \frac{M}{M_{ref}}$	16 Gb/s	16 Gb/s	100 Gb/s	80 Gb/s
6	$(PR + CR) \cdot \frac{BW}{BW_{ref}} \cdot \frac{N_s}{N_{s_{ref}}} \cdot \frac{M}{M_{ref}}$	16.53 Gb/s	30.08 Gb/s	103.33 Gb/s	150.40 Gb/s
7.3 (only DL)	$(PR + CR) \cdot \frac{BW}{BW_{ref}} \cdot \frac{N_s}{N_{s_{ref}}} \cdot \frac{M}{M_{ref}} \cdot \frac{1}{R} + MAC_{info}$	50.4 Gb/s	—	310.80 Gb/s	—
7.2	$N_{SC} \cdot N_{sym} \cdot W \cdot N_s \cdot 1000 + MAC_{info}$	86.71 Gb/s	86.13 Gb/s	430.78 Gb/s	430.20 Gb/s
7.1	$N_{SC} \cdot N_{sym} \cdot W \cdot N_{AP} \cdot 1000 + MAC_{info}$	688.25 Gb/s	688.21 Gb/s	6.88 Tbps	6.88 Tb/s
8	$SR \cdot W \cdot N_{AP} \cdot 5$	1.25 Tb/s	1.25 Tb/s	12.58 Tb/s	12.58 Tb/s

TABLE 2. Equations to calculate the transport capacity for different functional splits and maximum fronthaul/midhaul capacity requirements for disaggregated 6G basic and advanced network scenarios, in DL and UL, considering the assumptions in Table 1.

ers are centralized. For split 1, the transport network capacity needs to support RRC signaling and user plane data. Compared with user plane data, RRC signaling is negligible. Then, the capacity is computed considering the data Peak Rate (PR), that is, the maximum user data rate to be supported from a system requirement point of view [3]. For 4G, the data PR is 150 Mb/s for DL and 50 Mb/s for UL. For 6G, the 4G data PR is scaled based on BW, modulation order (M), and number of MIMO streams (N_s). In the case of split 2, extra signaling (S_{PDCP}) is transmitted between the transport network, therefore the needed capacity is incremented with this extra signaling. Then, for splits 2 and 4, the capacity is equal to option 1, and for the case of split 2, the corresponding extra signaling is added to the capacity.

Split 6, in addition to RRC, PDCP and RLC, also centralizes the MAC layer. At the same time, the PHY and RF layers are implemented in the DU/RU. Therefore, from this split onwards, the radio resource scheduling is done in a centralized way. Therefore, split 6 adds to this equation the control and scheduling signaling rate (CR) that is transmitted from the MAC layer to the PHY layer and vice versa. This information contains the radio resource scheduling and HARQ process NACK/ACK response, among other information.

Split option 7 and its subvariants, that is, splits 7.x, divide the PHY layer internally. Split 7.3 centralizes the channel coding process belonging to the PHY layer [9]. This centralization only affects DL transmissions, as the DL channel coding is implemented in the CU/DU, while the rest of the PHY functions are located in the RU [3]. Therefore, the transport network must be able to transmit the codified packets or codewords, thus increasing the required capacity. Apart from this, additional MAC information (MAC_{info}) has to be transmitted to the low-PHY layer to be able to support modulation, multi-antenna processing, and radio resource allocation [11].

The calculation of capacity undergoes significant changes from split 7.3 to split 7.2 since in this case the modulation and spatial streams mapping are applied in the CU/DU. Consequently, the transport network capacity requirement increases significantly with split 7.2. Considering a bandwidth that corresponds to N_{SC} subcarriers in one OFDM symbol, and the transmission of data is carried out in N_{sym} OFDM symbols (during 1 ms), using W bits to represent each sample and the availability of multiple spatial streams (N_s) reusing the same subcarriers and symbols, then, the capacity is calculated as shown in Table 2. In Split 7.1, the antenna ports are centralized. Until now, transport capacity has been calculated by scaling the MIMO streams, however, for split options 7.1 and 8, the transport capacity is scaled based on the number of antenna ports N_{AP} .

Finally, split 8 locates the RF in the RU, and the rest of the layers are implemented in the CU/DU. Therefore, the RU only processes the RF signal and applies the analog-to-digital or digital-to-analog conversion. Then, the transport needs to transmit IQ samples to the DU/CU [9], increasing the needed capacity, being the most restrictive capacity requirement among the different functional split options.

Considering the equations from Table 2 (second column), we have calculated the maximum transport capacity that the fronthaul and midhaul networks need to support depending on the selected functional split and the assumed 6G scenario. The results for single-band are shown in Table 2 (third to sixth columns). W is 32 b/sample and the MAC information and additional signaling overheads follow [3]. In multi-band, the BW and sampling rate have to be multiplied by the corresponding CC number, and so does the corresponding transport capacity.

Let us note that the expressions/values in Table 2 do not consider fronthaul compression. However, the capacity expressions/values can be reduced when fronthaul compression techniques

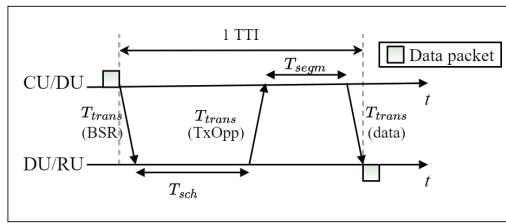


FIGURE 1. RLC-MAC interaction with split 4 configuration.

are applied [12]. For example, O-RAN has adopted point-to-point compression by means of block compression techniques (including block floating point, block scaling, and μ -law) and modulation compression for user data. For a typical system configuration, block compression techniques achieve a compression ratio of 0.58, while modulation compression gets a compression ratio of 0.25 (for 256-QAM) and 0.18 (for 64-QAM) [12], thus reducing accordingly the required transport capacities. However, modulation compression just applies to split option 7.2, while block compression techniques can be used for other functional splits.

To address such high transport capacity requirements, innovative ultra-high-speed, low-latency, low-cost, and power-efficient transceivers are developed for 6G fronthaul and midhaul networks [6]. Commercial optical technologies such as Ultrawideband Wavelength Division Multiplexing (UWB WDM), Spatial Division Multiplexing (SDM), and Subcarrier Multiplexing (SCM) enable to provide spatially-diverse point-to-multi-point optical distribution networks for 6G Xhaul. This will reduce the cost and power consumption of the fronthaul network, enabling an early adoption by mobile operators.

FRONTHAUL/MIDHAUL LATENCY REQUIREMENTS

The transport latency requirement (T_{trans} , which is equal to T_{FH} for fronthaul and T_{MH} for midhaul) is calculated considering the interactions occurring between the protocol layers that are included in CU and DU, or DU and RU. The latency requirement depends on the functional split, and it is determined mostly by the most restrictive process in 5G/6G network that requires signal exchange between different units.

The transport latency requirement in 5G is divided into three groups. The first group consists of split 1 and split 2, where only high protocol layers RRC and PDCP are centralized. In this case, the latency requirement is constrained by network configuration, which happens every tens of ms, and so it is not very restrictive. The second group concerns Split 4, and the third group encompasses low-layer splits including options 6, 7 (and its subvariants), and 8.

Split 4 divides the RRC/PDCP/RLC in the CU/DU pool, while MAC/PHY/RF layers are stored in different RUs. The most restrictive iteration concerning latency is between the RLC layer and the MAC layer, as the scheduling between these layers is done per Transmission Time Interval (TTI). Each time a packet arrives from the data network, it is stored in the RLC buffer, and the MAC layer is responsible for scheduling the time and frequency of radio resources that are going to be used to transmit the packet. This process follows three steps (Fig. 1). First, the RLC layer sends information about the stored amount of

data (i.e., the buffer status report) to the MAC layer through the transport. Based on the quality-of-service requirements and the amount of data the RLC buffer contains, the MAC layer schedules radio resources to transmit this data. Then, the MAC layer informs the RLC layer about the transmission opportunity (TxOpp). This information is sent through the transport again. After receiving TxOpp information, RLC creates Packet Data Units to forward them to the MAC layer, entailing a processing time to segment a packet or prepare the multiple packets to be concatenated depending on the allocated TxOpp. Figure 1 shows the interaction between RLC and MAC layers.

Following Fig. 1 we can calculate the required transport latency for split 4. Assuming 1 ms TTI [3], 500 μ s MAC scheduling time, and 200 μ s packet segmentation time, then $T_{trans} = 100 \mu$ s is the maximum delay that the transport network has to support to implement split 4.

With the splits 6/7.x/8, the MAC layer processes are done in a centralized way and the decisions made by the MAC layer, such as the radio resource scheduling for each packet, are exchanged with the DU or RU through the transport network. The most latency-restrictive process in the MAC is the HARQ process [3]. In 4G, the HARQ process is fixed to 4 ms, that is, gNB has to send data and receive the corresponding ACK/NACK feedback within 4 ms. In this context, the gNB transmits the packet to the UE over the air. This time can be affected by different technologies. If we consider a 6G network with carrier aggregation, the propagation time in the RAN will be equal to the time it takes for the packet to travel over the air since it occurs in a single hop. However, if an NR repeater or a UE-to-network relay is required, the total time is the sum of three processes, including signal transmission to the intermediate node over the air, processing at the repeater/relay, and signal retransmission from the repeater to the UE over the air. This is further extended if multiple hops are considered. Additionally, if different RUs are used for transmitting and receiving signals through dual connectivity or coordinated multi-TRP, the propagation times will vary depending on which RU is being used. Consequently, these technologies will add latency to the overall t_{prop}^{RAN} time shown in Fig. 2. After sending the packet to the UE, the packet is decoded, and the UE has to process the data and prepare the ACK/NACK response. The response is transmitted again to the gNB through the air. In 5G, the maximum time the HARQ process can take is determined by $k1$ variable which represents the time between the slot where DL data is transmitted (gNB to UE) and the slot where ACK/NACK message is received at the gNB. The $k1$ value is selected depending on the numerology and the UE processing capabilities. In case DLUL decoupling technology is used, the UL RAN propagation time can be different from the DL one. Therefore, in [8], it is mentioned that increasing the HARQ time and implementing this mechanism in the control base station is a possible solution for ensuring proper HARQ functionality in a fully decoupled RAN architecture. However, if HARQ is not used for error control and retransmissions are managed by higher layers [8], the transport latency requirements will be relaxed.

To calculate the transport latency for split 6, we refer to Fig. 2. We assume numerology 0 and a value of $k1 = 4$ ms, as in 4G, while considering negligible propagation delay (assuming a single hop between the RU and the UE). We also consider the time needed for decoding and ACK/NACK preparation in the UE is equal to $500 \mu\text{s} + 2$ TTIs. Based on these parameters, the transport latency for split 6 and beyond is given by $T_{trans} = 250 \mu\text{s}$. This latency depends on the TTI, which can vary based on the numerology or subcarrier spacing (SCS) employed in the 5G NR system. 5G NR already considers TTIs of 1 ms, $500 \mu\text{s}$, $250 \mu\text{s}$, and $125 \mu\text{s}$ for data channels that correspond to SCS = {15, 30, 60, 120} kHz. Even if it is not clear what will happen in 6G, probably these SCSs are going to be supported. So, the latency requirements for 6G transport networks can be extrapolated from 5G, considering different TTI durations. For split 4, when SCS = 60 kHz (numerology 2), $T_{trans} = 25 \mu\text{s}$ and for split 6 onwards, $T_{trans} = 62.5 \mu\text{s}$.

As we have stated before, depending on which unit the protocol layers are implemented, T_{trans} will be equal to T_{FH} or T_{MH} . Therefore, if split 2 is used for midhaul and split 6 for fronthaul, then the previously derived requirements apply, respectively, to each network segment, that is, 1.5–10 ms for midhaul and $250 \mu\text{s}$ for fronthaul. However, there is a very particular case that needs special attention. In case functional splits 6/7.x/8 are adopted for both midhaul and fronthaul networks (e.g., split 6 for CU/DU and split 8 for DU/RU), then the derived latency requirements need to be shared across these two transport networks. That is, the $250 \mu\text{s}$ requirement needs to be satisfied by summing the $T_{MH} + T_{FH}$ delay, which is even more restrictive. Figure 2 shows an example for this particular case.

FRONTHAUL/MIDHAUL TRANSPORT NETWORK SCENARIOS FOR 6G USE CASES

In this section, we provide a general overview of the architecture that the fronthaul and midhaul transport networks must support to make feasible two use cases: dynamic FFSs, and cell-free massive MIMO connectivity. Furthermore, we calculate the transport network requirements for these use cases when considering a 6G network. Specifically, we extrapolate the values derived earlier for a simple scenario to a multi-RU and multi-DU scenario and show how these values can increase significantly.

USE CASE 1: DYNAMIC FLEXIBLE FUNCTIONAL SPLITS

This scenario aims to realize the potential of dynamic FFS as a novel RAN alternative going one step beyond the fixed functional split adopted by 3GPP and O-RAN. Centralizing baseband processing reduces power consumption [5]; however, a fully centralized network with a fixed split implies highly restrictive latency and throughput requirements on the fronthaul. To address this, the key concept is to enable different functional split options at the RU/DU/CU and dynamically adjust the functional split selections to adapt the 6G network to varying environment conditions (new services/use cases, traffic changes, etc.). This approach aims to centralize baseband processing when possible, thus reducing power consumption. Furthermore, as mentioned in [5], total costs for operators (including migration

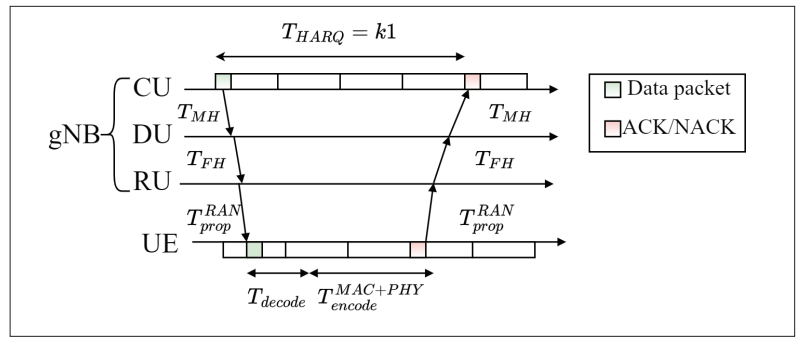


FIGURE 2. Transport network latency requirement considering split 6 between CU-DU and splits 7.x or 8 between DU-RU.

and operating costs, the latter considering computational, deployment, and routing costs) can be reduced by jointly optimizing the functional split and activating/ deactivating small cells RUs and DUs, while centralizing the protocol layers in the CU when feasible. Furthermore, FFS is a technology that promotes a user-centric network, as it can select different functional splits for each RU based on the available traffic at any given moment, with the flexibility to adjust the selected functional split option as user requirements change [5]. To this end, the targeted scenario assumes the deployment of regional Central Office (CO)s equipped with a metro datacenter, providing a CU pool and a metro router to aggregate traffic from different access COs. The access CO itself consists of an edge computing datacenter deploying a DU pool, along with an aggregation router combining traffic from multiple cell sites. At the cell sites, RUs are connected to a Cell Site Gateway (CSG) router that provides connectivity to the far-edge computing data center. DUs and CUs are dynamically deployed as virtual network functions in the computing servers located in the access and regional central offices respectively. As for the RUs, the lowest processing of the PHY layer is performed by hardware (e.g., Field Programmable Gate Array (FPGA)), but the other radio processing functions are deployed as software instances in computing servers. A single DU placed in the access central office aggregates traffic from a set of RUs connected in a logical tree topology. Then a single CU located in a regional central office aggregates traffic from a set of DUs connected also in a logical tree topology.

Figure 3 shows an example of a 6G transport network scenario for dynamic FFS. It considers various degrees of functional splits in the RU/DU/CU, ranging from the most restrictive (blue or RU_1) functional split in terms of fronthaul network capacity and latency requirements to the less demanding (orange or RU_2). RU_1 adopts the functional splits specified by O-RAN, with the RU hosting only the RF and low PHY functions, the DU handling high PHY, MAC and RLC functions, and the CU responsible for the PDCP. This represents a fronthaul split option 7.2 and midhaul split option 2. For RU_2 , the PHY and MAC functions are handled at the RU, the RLC functions at the DU, and PDCP at the CU, representing fronthaul split option 4 and midhaul split option 2. Following Table 2, for functional split option 7.2, the transport capacity requirements, accounting for both the 6G basic and advanced scenarios, ranges from 86 to 430 Gb/s both

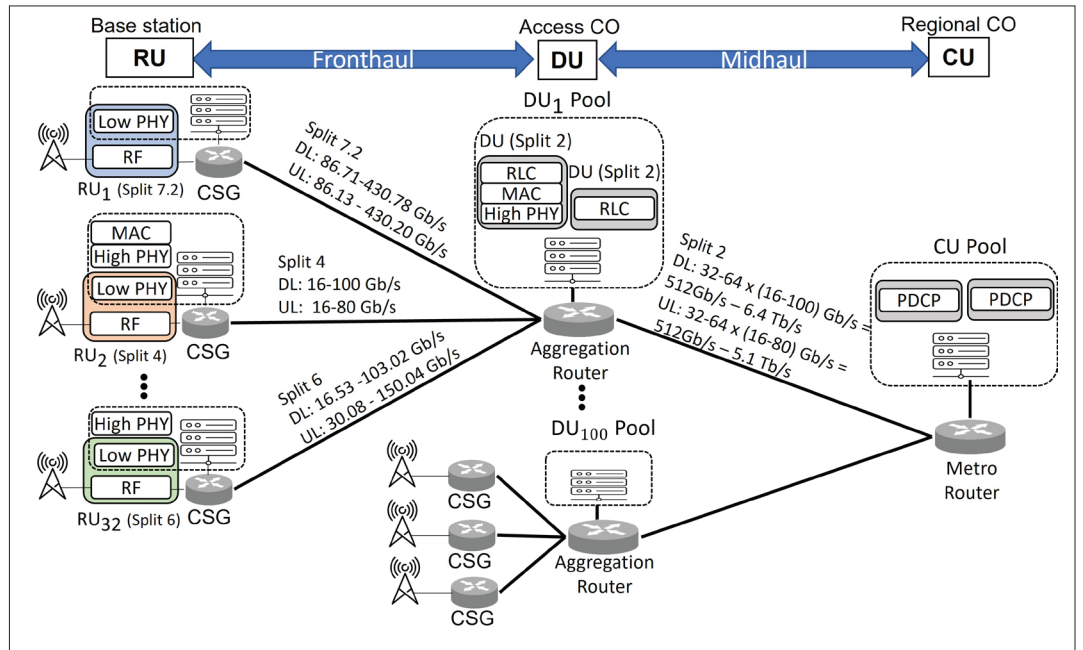


FIGURE 3. 6G transport network scenario for dynamic FFS.

for DL and the UL. While, for split 2 and 4, the transport capacity requirements, accounting for both the 6G basic and advanced scenarios are 16–100 Gb/s and 16–80 Gb/s for the DL/UL transport networks. Moreover, considering a single DU can process traffic from 32 up to 64 RUs [4], then, the transport capacity requirements of the midhaul transport segment connecting the DU and the CU ranges from 512 Gb/s to 6.4 Tb/s, based on the number of aggregated RUs and considering the functional split option 2 in the DL path. Finally, a single CU can process over 100 DUs [4]. Therefore, the back-haul capacity requirement between the CU and the core network can range from 51.2 Tb/s to 640 Tb/s.

With FFS, different functional split options can be selected for each of the cells by adapting the fronthaul/midhaul connections to the required transport capacity according to the specific demands of new services, traffic changes, and so on. For example, we will suppose that the orange or RU_2 cell site that was previously configured with functional split option 4, is now reconfigured with split option 7.3 for the RU/DU split. This requires increasing the fronthaul transport capacity, from 16–100 Gb/s to 50.4–310.80 Gb/s for the DL, and from 16–80 Gb/s to 50.4–310.80 Gb/s for the UL. However, this allows performing joint processing of the blue and orange cells MAC functions in a single DU pool.

USE CASE 2: CELL-FREE MIMO

The second scenario is designed to support cell-free massive MIMO connectivity [13]. In this case, the notion of cell disappears and the UE signal is transmitted/received and processed by multiple RUs (N_{RU}), where $N_{RU} > 2$ spatially distributed antenna arrays are used to perform distributed MIMO. The overall transport network scenario in this cell-free massive MIMO is shown in Fig. 4. The entire radio signal processing for multiple RUs may now be performed at an access CO, enabling advanced MIMO processing at centralized loca-

tions (e.g., coordinated multi-point joint transmission, coordinated beamforming, or coordinated scheduling), enhancing radio signal performance in terms of users, capacity, and coverage. To enable such advanced centralized processing, this scenario adopts functional split option 8 for the fronthaul interface, whereby RUs only host the RF functions, while access COs comprise Central Processing Unit (CPU)s hosting the full set of radio functions covering from PHY to PDCP.

To address this use case, there are two main transport connectivity needs. The first one that fronthaul connections between the antenna sites and the access COs are connected in a logical tree topology, while the second is for intra-CPU communication between access COs are connected in a physical ring topology, for coordination and/or joint processing between CPUs. At the access COs there is an aggregation router that serves to concentrate traffic from the distributed antenna sites and enable communication with other CPUs. The peak transport capacity requirements, accounting for both the 6G basic and 6G advanced scenarios are 1.25–6.29 Tb/s for each DL/UL fronthaul connection.

CONCLUSIONS

This article has derived the capacity and latency requirements that transport networks must meet to operate effectively with different functional splits in a 6G disaggregated RAN. These requirements were derived based on various envisioned 6G network scenario configurations while considering emerging 6G technologies. Additionally, we proposed specific fronthaul/midhaul transport network scenarios to address the imposed requirements in two 6G use cases: flexible functional split and cell-free MIMO. This work will serve to set the basic requirements for the optical network research and development community, in order to design, implement and optimize new optical transceivers and packet-optical network architectures to meet the future disaggregated RAN demands.

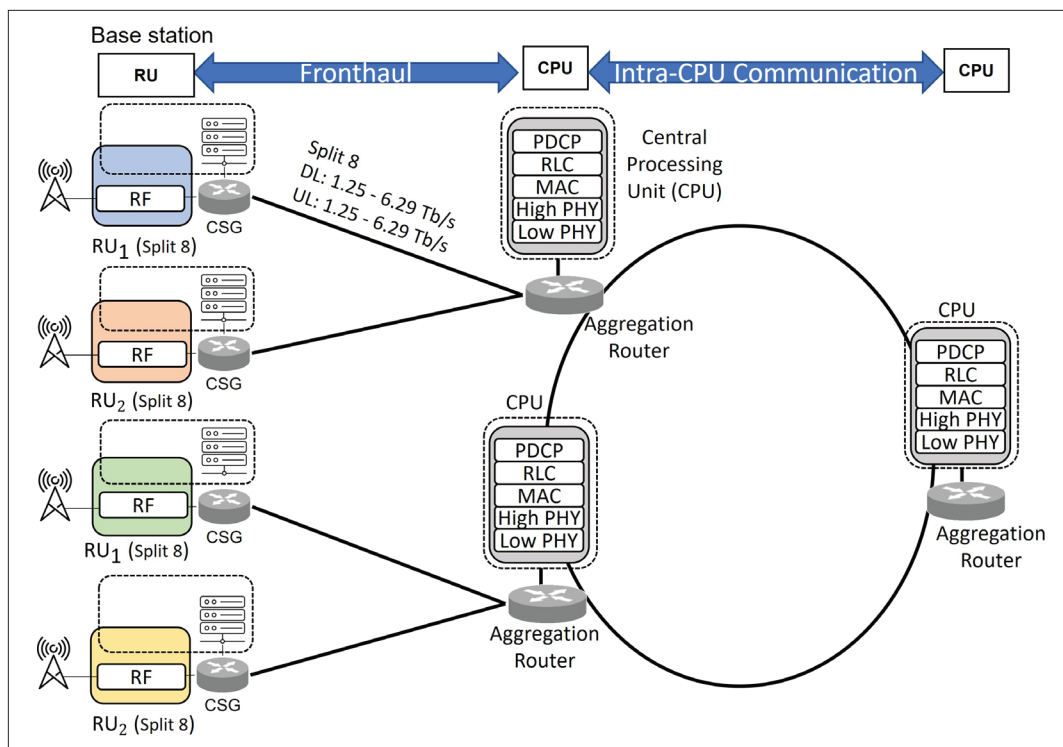


FIGURE 4. 6G transport network scenario for cell-free MIMO.

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